Experiment – 6 Parseval's Relation and Harmonic Spectrum Analysis

Objective 1:

To verify Parseval's relation for continuous-time periodic signals and analyze the relationship between time-domain and frequency-domain power

Experimental Procedure:

• Generate the periodic signal over one period.

• Compute time-domain average power.

• Calculate Fourier coefficients up to N = 20.

• Evaluate frequency-domain power using Parseval's relation.

• Compare the two values and calculate absolute error.

Code:

```
clc;
clear;
close all;

% Verification of Parseval's Relation
f0 = 1;
T = 1/f0;
t = linspace(0,T,4001);

% Half-wave rectified sinusoidal signal
x = max(sin(2*pi*f0*t),0);

% Time-domain average power
P_time = (1/T)*trapz(t,abs(x).^2);

% Fourier Series coefficients
```

```
N = 20;
a0 = (2/T)*trapz(t,x);

an = zeros(1,N);
bn = zeros(1,N);

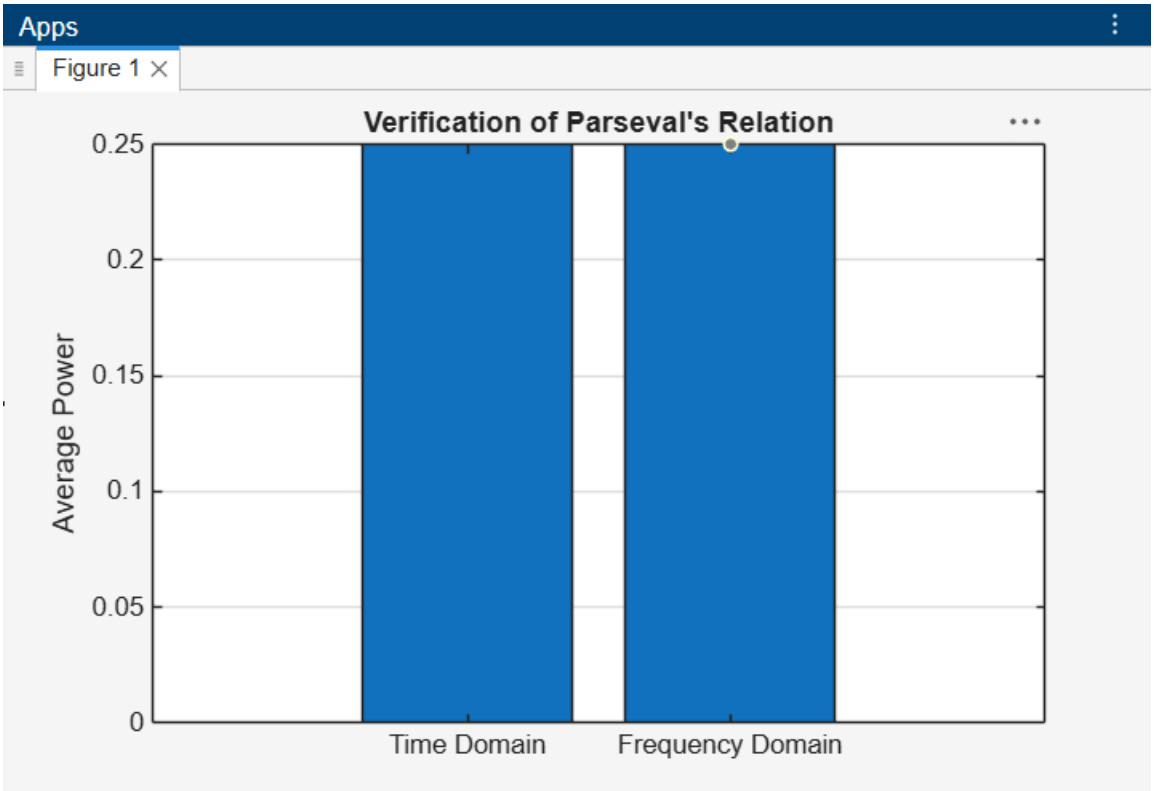
for k = 1:N
   an(k) = (2/T)*trapz(t,x.*cos(2*pi*k*f0*t));
   bn(k) = (2/T)*trapz(t,x.*sin(2*pi*k*f0*t));
end

% Frequency-domain average power
P_freq = a0^2/4 + 0.5*sum(an.^2 + bn.^2);

% Display results
fprintf('Time-domain power     = %.6f\n',P_time);
fprintf('Frequency-domain power = %.6f\n',P_freq);
fprintf('Absolute error        = %.6e\n',abs(P_time-P_freq));

% Plotting
figure;
bar([P_time P_freq]);
set(gca,'XTickLabel',{'Time Domain','Frequency Domain'});
ylabel('Average Power');
title('Verification of Parseval''s Relation');
grid on;
```

output:



Objective 4:

To compare the total power obtained in the time domain with the harmonic power distribution in the frequency domain.

Experimental Procedure:

- Generate the periodic signal and compute Fourier coefficients.
- Calculate harmonic magnitude for n = 1 to 15.
- Convert harmonic number to frequency.
- Plot the harmonic spectrum using a stem plot.
- Identify the significant components

Code:

```
clc;
clear;
close all;

% Time-Domain and Frequency-Domain Power Comparison
```

```
f0 = 1;
T = 1/f0;
t = linspace(0,T,4001);

% Half-wave rectified sinusoidal signal
x = max(sin(2*pi*f0*t),0);

% Time-domain power
P_time = (1/T)*trapz(t,x.^2);

% Fourier coefficients
N = 20;
a0 = (2/T)*trapz(t,x);

an = zeros(1,N);
bn = zeros(1,N);

for k = 1:N
   an(k) = (2/T)*trapz(t,x.*cos(2*pi*k*f0*t));
   bn(k) = (2/T)*trapz(t,x.*sin(2*pi*k*f0*t));
end

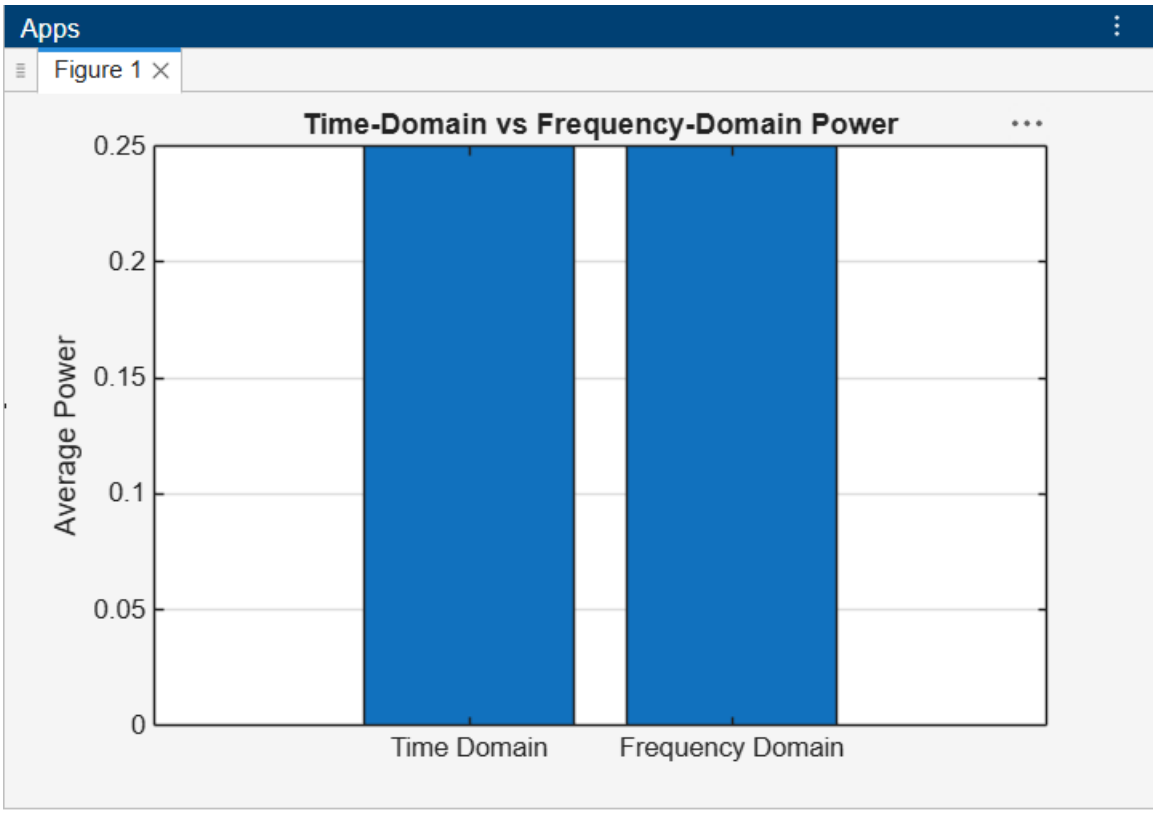
% Frequency-domain power
P_dc = a0^2/4;
P_harmonics = 0.5*sum(an.^2 + bn.^2);
P_frequency = P_dc + P_harmonics;

% Display results
```

```
fprintf('Time-domain power     = %.6f\n',P_time);
fprintf('DC power              = %.6f\n',P_dc);
fprintf('Harmonic power        = %.6f\n',P_harmonics);
fprintf('Frequency-domain power = %.6f\n',P_frequency);

% Plotting
figure;
bar([P_time P_frequency]);
set(gca,'XTickLabel',{'Time Domain','Frequency Domain'});
ylabel('Average Power');
grid on;
title('Time-Domain vs Frequency-Domain Power');
```

output:



Result:

Thus, Parseval's relation was successfully verified by calculating and comparing the time-domain and frequency-domain average power of periodic signals. The DC power

and harmonic power were also analyzed, and both methods produced approximately equal power values.

Objective	**Assessment Criterion**	**Excellent – 4**	**Good – 3**	**Satisfactory – 2**	**Needs Improvement – 1**	**Weightage**	**Marks (10)**	**Marks obtained**
(i) To verify Parseval's relation for continuous-time periodic signals and analyze the relationship between time-domain and frequency-domain power.	**Fourier Series Coefficient Computation**	Correctly identifies the fundamental period T0 and frequency ω0 of the periodic signal. Computes the Fourier series coefficients (c_k, or a_k and b_k) accurately, analytically or in MATLAB, and plots the signal with its harmonic reconstruction with correct labels and scaling.	Computes the coefficients correctly with minor errors in the period, limits, scaling or labelling.	Computes only some coefficients correctly, or has noticeable errors in T0, integration or code, and needs guidance.	Unable to determine the period or compute the Fourier series coefficients.	**20**	2	

	Time-Domain Power & Parseval Verification	Correctly computes the average power P .Applies Parseval's relation correctly, verifies that the two values agree within a small numerical error, and explains the time-frequency relationship clearly.	Computes the power and applies Parseval's relation correctly, with minor numerical or interpretation errors.	Computes the time-domain power or applies Parseval's relation only partly. Verification is incomplete or has noticeable errors, and needs guidance.	Unable to compute the power or verify Parseval's relation.	**20**	2	
(ii) To compare the total power obtained in the time domain with the harmonic power distribution in the frequency domain.	**Harmonic Power Distribution**	Correctly computes the power in each harmonic and the DC power. Plots the power spectrum (stem plot) with labelled axes, shows the cumulative power, and identifies the dominant harmonics accurately.	Computes the harmonic powers and plots the spectrum correctly, with minor errors in indexing, scaling or labelling.	Computes some harmonic powers but has noticeable errors in the spectrum or the DC and harmonic terms, and needs assistance.	Unable to compute or plot the harmonic power distribution.	**20**	2	

	Time vs Frequency Power Comparison	Compares total time-domain power with the sum of harmonic powers for an increasing number of harmonics. Reports the percentage error and the power captured, explains convergence and the effect of truncation (for example for square or triangular waves), and draws correct conclusions.	Compares the total powers correctly, with minor errors in the error calculation or limited explanation of convergence.	Makes a basic comparison with incomplete error analysis, or gives limited explanation, and needs guidance.	Unable to compare the time-domain and frequency-domain power correctly.	**20**	2	
Interpret the results for both objectives and maintain a proper technical record.	**Engineering Analysis, Record & Interpretation**	Provides technically accurate interpretation of the results, relating power to harmonic content and signal waveform.	Maintains a proper record and gives mostly accurate interpretation, with minor omissions in explanati	Record and interpretation are basic, with partial explanation or missing elements.	Incomplete record, limited or incorrect interpretation of the results.	**20**	2	

		Maintains a complete record with aim, theory, MATLAB code, plots, observations, result and conclusion, with clear organization.	on or presentation.					
					Total	**100**	**10**	